PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Are COVID-19 models blind to the social determinants of health? A systematic review protocol
AUTHORS	John-Baptiste, Ava; Moulin, Marc; Ali, Shehzad

VERSION 1 – REVIEW

REVIEWER	Santosh, K. C. University of South Dakota
REVIEW RETURNED	14-Feb-2021

GENERAL COMMENTS	# Interesting topic # The following are my comments: + Limited originality + No well articulated objectives are presented. + Writing: good, but it requires serious proofread. + References: avoid non-peer-reviewed articles, such a pre-print and arxiv. They are no more than a repository. # Comparison is missing.

REVIEWER	Walker, Blake Friedrich-Alexander-Universität Erlangen-Nürnberg
REVIEW RETURNED	31-Mar-2021

GENERAL COMMENTS	Thank you kindly for the opportunity to review this protocol. This is a strong, well-justified review design and suitably comprehensive for the topic area. The protocol appears to closely follow PRISMA and is very low-risk, nor are there any ethical concerns. My only recommendation: Might the authors consider including an explicit built environment/infrastructure category in the review protocol? There have been several studies examining how these features exhibit an effect on COVID-19 incidence and exhibit interaction with socioeconomic risk factors. It might be worthwhile orientating the reader with one or two of these studies in the protocol, e.g. Scarpone et al. 2020 in IJHG. Interaction effects may play an important role in understanding socioeconomic, spatial, and temporal patterns of reported incidence, e.g., the positive effect of intercity rail connectivity appears to be stronger for medium- and high-income populations, whereas the negative effect of youth unemployment appears to be

	stronger among low-income populations. One minor point: the timeline is unclear. At what time do the authors expect the search to end (i.e., latest date in the literature search - the authors stat "to present") and at what time do the authors expect to submit their results for peer-review? The timeline can be quite important when examining a current outbreak, as it helps the reader to interpret the results in light of, e.g., a recent "wave". Aside from this, I have no additional recommendations or comments. The manuscript is well-written, clear, and scientifically sound. Thank you for the opportunity to reivew this work; I look forward to seeing the study results.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1
Comments to the Author:

-Interesting topic
-The following are my comments:
-Limited originality
-No well articulated objectives are presented.
-Writing: good, but it requires serious proofread.
-References: avoid non-peer-reviewed articles, such a pre-print and arxiv. They are no more than a repository.
-Comparison is missing.

RESPONSE

- We thank Reviewer 1 for valuable feedback on our manuscript. Given the lack of exploration of the incorporation of social determinants of health into COVID-19 infectious disease models, we feel that this work is original and valuable. We feel the systematic review provides an important opportunity to critically appraise modelling efforts with respect to social determinants of health.

-We stated the objectives in the last paragraph of the introduction section. In response to the reviewer's comments, we have revised the text to ensure the statement of objectives is clearer. In particular, we have reworded the text to include an explicit objectives statement.

-The document has been proofread and changes to the text have been made throughout the manuscript.

-We thank the reviewer for comments about preprint archives. We will be removing preprint and arxiv articles from the systematic review. Only published studies will be included.

- In response to the reviewer's comments about comparison, we have added to our manuscript a discussion of our goals to (1) compare models in terms of the approaches used to incorporate social factors; (2) conduct a temporal comparison to identify trends and see if an increasing number of published models have incorporated social factors over time; and (3) based on our knowledge of published epidemiological estimates that quantify the association between social factors and COVID-19, compare incorporation of social factors in COVID-19 models to this evidence base (Evidence synthesis section).

Reviewer: 2
Comments to the Author:

Thank you kindly for the opportunity to review this protocol. This is a strong, well-justified review design and suitably comprehensive for the topic area. The protocol appears to closely follow PRISMA and is very low-risk, nor are there any ethical concerns.

My only recommendation: Might the authors consider including an explicit built environment/infrastructure category in the review protocol? There have been several studies examining how these features exhibit an effect on COVID-19 incidence and exhibit interaction with socioeconomic risk factors. It might be worthwhile orientating the reader with one or two of these studies in the protocol, e.g. Scarpone et al. 2020 in IJHG.

Interaction effects may play an important role in understanding socioeconomic, spatial, and temporal patterns of reported incidence, e.g., the positive effect of intercity rail connectivity appears to be stronger for medium- and high-income populations, whereas the negative effect of youth unemployment appears to be stronger among low-income populations.

One minor point: the timeline is unclear. At what time do the authors expect the search to end (i.e., latest date in the literature search - the authors stat "to present") and at what time do the authors expect to submit their results for peer-review? The timeline can be quite important when examining a current outbreak, as it helps the reader to interpret the results in light of, e.g., a recent "wave".

Aside from this, I have no additional recommendations or comments. The manuscript is well-written, clear, and scientifically sound. Thank you for the opportunity to review this work; I look forward to seeing the study results.

RESPONSE

We thank Reviewer 2 for the extremely helpful comments on ways to improve the manuscript.

-We considered an explicit built environment/infrastructure category in the review protocol, but feel that while built environment is an important consideration, it does not directly represent a social factor. We think that any potential influence of built environment in the context of the current review is likely to be captured by rurality/urbanity, population density and the sociodemographic factors included in the review. As such, we feel that this category is outside the scope of the proposed

systematic review.

-A timeline for start and end dates as well as when we expect to submit for peer-review publication has been added to the methods section.

Once again, we are grateful for the reviewers' comments, which we feel have allowed us to improve the manuscript significantly.

VERSION 2 – REVIEW

REVIEWER	Walker, Blake Friedrich-Alexander-Universität Erlangen-Nürnberg
REVIEW RETURNED	18-May-2021

GENERAL COMMENTS	The authors have adequately addressed all reviewer concerns.